

13.05 Arithmetic Series

Common problems and solution methods

Rule guide: recognize the task first

Recognize	Apply these rules
Sequences And Series	Check whether the pattern is additive or multiplicative before choosing a formula.
The expression contains the chapter's key notation or model.	Write the known quantities, choose the matching formula, calculate carefully, and check the result in context.

Example 1: Identify the model

Problem

$$a_{10} = 5 + (10 - 1)3$$

How to solve it

Write the known quantities and restrictions.

$$a_n = a_1 + (n - 1)d$$

Check the result against the original statement.

Example 2: Apply the key rule

Problem

$$a_n = 4(2)^{n-1}$$

How to solve it

Write the known quantities and restrictions.

$$a_n = a_1 r^{n-1}$$

Check the result against the original statement.

Example 3: Check the result

Problem

$$S_{20} = \frac{20(4 + 61)}{2}$$

How to solve it

Write the known quantities and restrictions.

$$S_n = \frac{n(a_1 + a_n)}{2}$$

Check the result against the original statement.